

FASTAPI BASICS

INSTALLATION

- `pip install fastapi uvicorn`
- `pip install "fastapi[all]"`
- Includes pydantic, jinja2, etc.

RUNNING THE SERVER

```
uvicorn main:app --host 0.0
--port 8000
Access at http://127.0.0.1:8000
```

AUTOMATIC DOCS

- `/docs` (Swagger UI)
- (Redoc)
- OpenAPI standard

CREATING APP

- `from fastapi import FastAPI`
`app = FastAPI()`
`uvicorn main:app --reload`

PATH PARAMETERS

- `uvicorn main:app --host 0.0.0.0 --port 8000`

`/docs`

- Standard Python types
- Data validation
- OpenAPI standard

CREATING APP

- `@app.get('/')`
- `@app.post('/items/')`

PATH PARAMETERS

- `def read_item(item_id: int):`
`return {"q": str(item_id)}`
- Type hints for validation

REQUEST BODY

- `from pydantic import BaseModel`
`(BaseModel; name: str)`
`def create_item(item: Item): ...`

ROUTES & METHODS

- `@app.get('/')`
`put('/users/{user_id}')`

QUERY PARAMETERS

- `def read_items(q: str = None):`
`return {"q": q}`
- Default values make optional

TYPE HINTS

- Standard Python types
- Data validation
- API protection

INTERMEDIATE FASTAPI CONCEPTS

DEPENDENCY INJECTION

- Decoupled logic
Reusable components
- Testing dependencies
Handles lifecycles

SECURITY (OAuth2)

- OAuth2PasswordRequestForm
TokenUrl
- Bearer Token

ERROR HANDLING

- HTTPException
detail
- Bearer Token
Authentication flow

MIDDLEWARE

- Request/response processing
Custom headers

MIDDLEWARE

- Request/response processing
Custom headers
- CORS, GZIP, etc.

APIROUTER

- HTTPStatus = status codes
- Non idempotent
- Error_exception_handler

TESTING

- from fastapi import
BackgroundTasks
- @app.middleware

BACKGROUND TASKS

- task.add_task();
task.add_task()
Non-blocking operation
operation

MULTI-FILE APPS

- include_router

TESTING

- from fastapi.testclient import
TestClient
client.get(), post(),
put(), delete()
- Assertions

EVENTS

- app.on_event("startup")
shutdown
- File downloads
Assertions

CUSTOM RESPONSE CLASSES

- from fastapi.responses: HTMLResponse,
StreamingResponse,
StreamingResponse = File
JSONResponse (default)

ADVANCED FASTAPI CONCEPTS

DEPENDENCY INJECTION INJECTION OVERRIDE

- Testing with mocks
- Dynamic components
- Handles lifespan

CUSTOM MIDDLEWARE

- Functionality at app level
- Access to request.responses
- Custom headers

REQUEST BODY

- Testing with mocks
- Movable components
- @app.dependency-overrides

MIDDLEWARE

- Testing with mocks
- Single exception handler
- @app.middleware("http")

GRAPHQL

- strawberry/TranSition
- API encoding
- Type-safe APIs

FASTAPI SUBSCRIBERS

- Mounting other apps
- API versioning
- fastapi.FastAPISubRouter

GRAPHQL

- Non-blocking operations
- async def
- Offloading CPU-bound tasks (thread pools)

ASYNC/AWAIT PATTERNS

- Non-blocking operations
- async def
- Offloading endpoints (Request.background)

EXCEPTION HANDLERS

- Consistent error responses
- Handling CPU-bound tasks (Request.background)

WEBHOOKS

- from fastapi import Webhook

WEBHOOKS

- Gunicorn/Uvicorn cache
- Data limiting
- Session storage

PULSAR

- Gunicorn/Uvicorn workers
- Load balancing
- Session storage